Clustering Premier League Teams by Playing Style:
An Unsupervised Learning Analysis Using K-Means and GMM

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Abstract

This report investigates whether English Premier League (EPL) teams exhibit distinct, identifiable playing styles that can be discovered automatically from match statistics. Using squad-level per-90-minute data from the 2024–2025 season sourced from FBref, we apply two unsupervised learning algorithms—K-Means and Gaussian Mixture Models (GMM)—to cluster all 20 teams across 15 selected playing-style features. The Elbow method and Bayesian Information Criterion (BIC) both support $k = 3$ as the optimal number of clusters. K-Means identifies a dominant possession-based group (9 teams including Arsenal and Liverpool), a resilient mid-table group (7 teams), and a struggling group (4 teams containing all three relegated sides). GMM produces a sharper, more selective elite cluster of 6 teams. Both algorithms agree on a core elite group. We discuss the effect of small sample size on GMM's soft-assignment capability and outline directions for future work.

1 Introduction

The English Premier League is widely regarded as one of the most competitive football leagues in the world, featuring teams with markedly different playing philosophies and tactical setups. Quantifying these differences objectively is valuable for tactical analysts, scouts, and broadcasters seeking to understand team identity beyond league position alone.

Unsupervised machine learning offers a natural framework for this task: given a set of statistical descriptors of how each team plays, clustering algorithms can partition teams into groups with similar styles without requiring labelled outcome data. This is particularly well-suited to football analytics, where rich performance data is available but ground-truth style labels are subjective and inconsistent.

Hypothesis: EPL teams can be automatically grouped into a small number of distinct, interpretable playing style clusters using per-90-minute squad statistics.

This study uses two complementary algorithms—K-Means [1] and GMM [2]—to cluster all 20 EPL teams from the 2024–2025 season. We evaluate the number of clusters using the Elbow method and BIC, compare the cluster structures, and analyse the statistical profiles of each group. Principal Component Analysis (PCA) [4] is used exclusively for exploratory visualisation.

2 Literature Review

The application of clustering to sports analytics has grown substantially with the availability of rich event-level data. Romero Clavijo et al. applied cluster analysis to identify football team playing styles using positional and passing metrics, finding that k-means clustering produced interpretable groupings consistent with expert tactical knowledge [5]. Their work directly validates the use of squad-level statistics as a proxy for tactical identity, motivating our approach.

K-Means clustering, introduced by MacQueen [1], partitions data by iteratively minimising within-cluster inertia via hard cluster assignments. Despite its simplicity, it remains competitive with more complex methods for moderate-dimensional problems. Its sensitivity to initialisation is mitigated

by the k-means++ strategy and multiple random restarts.

Gaussian Mixture Models (GMM) offer a probabilistic alternative: each cluster is modelled as a multivariate Gaussian, and the Expectation-Maximisation (EM) algorithm [2] is used to fit the parameters. GMM's principal advantage is *soft* cluster assignments—each observation receives a posterior probability of belonging to each cluster—making it more informative than K-Means' hard boundaries. Model complexity is controlled through the BIC [3].

Bishop [3] provides a comprehensive treatment of both K-Means and GMM within a probabilistic framework. The EM algorithm for GMM is shown to be a special case of variational inference, and BIC is recommended as a principled criterion for model selection when the number of components is unknown.

PCA is employed here solely for 2D visualisation. As described by Jolliffe [4], PCA identifies orthogonal directions of maximum variance, enabling high-dimensional data to be inspected visually. Following standard practice, clustering is performed on the original scaled features rather than PCA-reduced representations.

Berrar et al. developed a data- and knowledge-driven framework for predicting soccer match outcomes using machine learning, demonstrating that integrating domain knowledge with statistical models yields robust predictive performance [6]. This motivates the use of principled, knowledge-informed machine learning methods in football analytics more broadly.

3 Data Processing

3.1 Data Source

Data were obtained from FBref [7], a publicly available football statistics platform. Four separate CSV files covering the 2024–2025 EPL season were downloaded: standard stats, shooting stats, miscellaneous stats, and goalkeeping stats. These were merged step-by-step on the `Squad` column, dropping duplicate columns introduced by each join:

Listing 1: Step-by-step dataset merging

```
master = standard.merge(
    shooting, on='Squad', suffixes=('', '
        _drop'))
master = master.drop(columns=[
    c for c in master.columns if c.endswith
        ('_drop')])

master = master.merge(
    misc, on='Squad', suffixes=('', '_drop'
        ))
master = master.drop(columns=[
    c for c in master.columns if c.endswith
        ('_drop')])

master = master.merge(
    goalkeeping, on='Squad', suffixes=('',
        '_drop'))
master = master.drop(columns=[
    c for c in master.columns
    if c.endswith('_drop') or c.endswith('
        .1')])
```

This produced a master dataset of 20 rows (one per team) and 47 columns:

Listing 2: Shape after merging

```
Shape: (20, 47)
```

All statistics are reported as per-90-minute averages. Since all 20 teams completed exactly 38 league matches, per-90 and per-match figures are directly comparable across teams.

3.2 Feature Selection

The 47 raw features were filtered in two stages to retain only those measuring playing *style* rather than team quality or match outcomes.

Stage 1 – Dropped categories:

- *Metadata:* `# Pl, Age, MP, Starts, Min, 90s` — scheduling constants or squad composition metrics.
- *Outcome metrics:* `Gls, Ast, G+A, G-PK, PK, PKatt, G+A-PK, G/Sh, G/SoT, GA, GA90, W, D, L, CS, CS%, PKwon, PKcon, PKA, PKsv, PKm, OG, 2CrdY` — these capture results, not the manner of play.

Stage 1 reduces the dataset to 17 style-related features:

Listing 3: Shape after Stage 1 feature drop

```
Shape: (20, 17)
['Poss', 'CrdY', 'CrdR', 'Sh', 'SoT', 'SoT
    %',
 'Sh/90', 'SoT/90', 'Fls', 'Fld', 'Off', '
     Crs',
 'Int', 'TklW', 'SoTA', 'Saves', 'Save%']
```

Stage 2 – Removed redundant features: Inspection of the correlation matrix revealed that `Sh/90` and `Sh` are nearly perfectly correlated ($r = 0.9999$), as are `SoT/90` and `SoT` ($r = 1.0$). This is expected since all teams played the same number of matches, making per-90 and total values proportional. Retaining both would give these features double weighting; the per-90 variants were dropped, leaving:

Listing 4: Shape after Stage 2 redundancy removal

```
Shape after removing redundant features:
    (20, 15)
['Poss', 'CrdY', 'CrdR', 'Sh', 'SoT', 'SoT
    %',
 'Fls', 'Fld', 'Off', 'Crs', 'Int', 'TklW',
 'SoTA', 'Saves', 'Save%']
```

The final feature set comprises 15 variables described in Table 1.

Table 1: Final 15 Features Used for Clustering

Feature	Category	Description
Poss	Possession	Ball possession (%)
Sh	Shooting	Shots per 90 min
SoT	Shooting	Shots on target per 90
SoT%	Shooting	Shot accuracy (%)
Crs	Crossing	Crosses per 90 min
Off	Set play	Offsides per 90 min
Fls	Discipline	Fouls committed per 90
Fld	Discipline	Fouls drawn per 90
CrdY	Discipline	Yellow cards per 90
CrdR	Discipline	Red cards per 90
Int	Defending	Interceptions per 90
TklW	Defending	Tackles won per 90
SoTA	Goalkeeping	Shots on target against
Saves	Goalkeeping	Saves per 90 min
Save%	Goalkeeping	Save percentage (%)

3.3 Preprocessing

No missing values were present in the dataset. Prior to clustering, all 15 features were standardised using `StandardScaler` (zero mean, unit standard deviation). This step is essential because K-Means and GMM are both distance- or covariance-based: features with larger numeric ranges (e.g. crosses per 90 ≈ 18) would otherwise dominate over features with smaller ranges (e.g. red cards per 90 ≈ 0.07). Standardisation ensures all features contribute equally. The first five rows of the scaled dataset are shown below (selected columns):

Listing 5: data_scaled.head() (selected columns)

```
Squad          Poss    CrdY    CrdR      Sh
    SoT
Arsenal        1.194  -0.901   2.451   0.670
    0.453
Aston Villa    0.087  -0.352   1.104  -0.114
    -0.180
Bournemouth   -0.259   1.536   0.296   1.039
    0.922
Brentford     -0.362  -1.622  -1.050  -0.668
    0.289
Brighton       0.399  -0.180   0.296   0.439
    0.630
```

PCA analysis shows that 6 components retain 86.7% of total variance (PC1: 33.3%, PC2: 15.6%, PC3: 12.2%, PC4: 10.3%, PC5: 8.3%, PC6: 7.0%), indicating moderate intrinsic dimensionality. The 2D PCA projection reveals an intuitive separation: high-possession sides (Arsenal, Liverpool, Manchester City) lie to the right, while lower-possession teams (Ipswich, Southampton, Leicester) appear to the left. Clustering is nonetheless performed on all 15 scaled features to avoid information loss.

```
In [5]: # Preview the first 5 rows to confirm the merge looks correct
        master.head()
```

Out[5]:

	Squad	# Pl	Age	Poss	MP	Starts	Min	90s	Gls	Ast	...	Sa
0	Arsenal	25	25.8	56.9	38	418	3,420	38.0	1.76	1.45	...	2
1	Aston Villa	28	27.0	50.5	38	418	3,420	38.0	1.47	1.18	...	2
2	Bournemouth	29	25.1	48.5	38	418	3,420	38.0	1.50	1.08	...	3
3	Brentford	28	25.8	47.9	38	418	3,420	38.0	1.71	1.16	...	4
4	Brighton	32	24.8	52.3	38	418	3,420	38.0	1.68	1.08	...	2

5 rows × 47 columns

Figure 1: Data Samples

4 Learning Methods

4.1 K-Means

K-Means [1] partitions n observations into k clusters by minimising the within-cluster sum of squares (inertia).

The algorithm alternates between assignment (each point to its nearest centroid) and update (recompute centroids) until convergence. We use

k-means++ initialisation with 10 random restarts (`n_init=10`) to avoid poor local minima.

Justification: K-Means is computationally efficient and produces interpretable centroids, making it a natural baseline for small datasets. Its assumption of spherical, equal-variance clusters is reasonable here given the modest problem scale.

4.2 Gaussian Mixture Models (GMM)

GMM [2] models the data as a weighted sum of k multivariate Gaussian distributions.

Justification: GMM allows ellipsoidal cluster shapes and provides uncertainty quantification via soft assignments. It is appropriate when cluster boundaries are not sharp, which may arise among mid-table teams with heterogeneous styles.

5 Analysis, Testing and Results

5.1 Model Selection

Fig. 2 shows the K-Means inertia and GMM BIC as functions of k. The inertia curve exhibits a clear elbow at k=3: inertia drops steeply from k=1 (300.0) to k=3 (192.9), then flattens. The BIC curve reaches its minimum at k=3 (−596.0), with k=4 close (−657.3) but requiring substantially more parameters. Both criteria consistently support k=3.

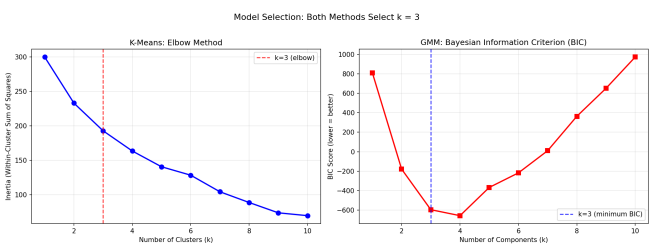

Figure 2: Model selection plots. *Left:* K-Means inertia shows an elbow at $k = 3$. *Right:* GMM BIC is minimised at $k = 3$.

5.2 K-Means Results ($k = 3$)

Table 2 summarises the K-Means clusters and key centroid values.

Attacking style (9 teams): The largest cluster captures possession-dominant, high-volume shooting teams. These sides average 54.9% possession and 14.4 shots per 90 minutes, and include all six teams that finished in the top eight. Their lower `TklW` (10.0) reflects that possession-based teams defend higher up the pitch, recovering the ball through pressing rather than deep tackling.

Table 2: K-Means Clustering Results ($k = 3$)

Style	Teams	Poss	Sh	TklW
Attacking (9)	Arsenal, Aston Villa, Brighton, Chelsea, Fulham, Liverpool, Man City, Newcastle, Spurs	54.9	14.4	10.0
Defensive (7)	Bournemouth, Brentford, Crystal Palace, Everton, Man Utd, Nott'm Forest, Wolves	46.1	12.5	11.4
Balanced (4)	Ipswich, Leicester, Southampton, West Ham	45.7	10.0	10.1

Defensive style (7 teams): These teams cede significant territory (46.1% possession) but generate a reasonable shot volume (12.5 per 90) and record the highest tackle rate of any cluster (TklW = 11.4). This profile is consistent with a low-block or counter-pressing approach: absorb pressure, win the ball back through defensive work rate, and attack on the break.

Balanced style (4 teams): The smallest cluster contains the three relegated sides (Ipswich, Leicester, Southampton) plus West Ham. These teams neither dominate possession (45.7%) nor generate many shots (10.0 per 90), and concede the most shots on target (SoTA = 5.59). Their profile reflects a passive, reactive style that was ultimately insufficient at this level.

5.3 GMM Results ($k = 3$)

GMM produces a tighter attacking cluster of 6 teams, separating Tottenham Hotspur and Brighton into the balanced group (possession of 50.6% but lower attacking output than the core elite). The defensive cluster (8 teams) is the largest, grouping together sides that cede possession and generate fewer shots, reflecting a more reactive playing style. The balanced cluster shows the highest tackle rate

Table 3: GMM Clustering Results ($k = 3$)

Style	Teams	Poss	Sh	TklW
Attacking (6)	Arsenal, Aston Villa, Chelsea, Liverpool, Man City, Newcastle	55.8	14.8	9.4
Balanced (6)	Brighton, Crystal Palace, Fulham, Man Utd, Spurs, Wolves	50.6	13.2	11.7
Defensive (8)	Bournemouth, Brentford, Everton, Ipswich, Leicester, Nott'm Forest, Southampton, West Ham	45.2	11.1	10.4

(TklW = 11.7), indicating active pressing despite only moderate possession.

5.4 Method Comparison

Both methods agree on a core *attacking* group: Arsenal, Aston Villa, Chelsea, Liverpool, Manchester City, and Newcastle United appear in the attacking cluster under both algorithms. This consistency across two methodologically distinct approaches strengthens confidence in this grouping.

The primary difference lies in the treatment of Tottenham and Brighton: K-Means assigns them to the attacking group based on their above-average possession, while GMM's flexible covariance structure places them in the balanced group given their lower attacking output relative to the core six.

A notable limitation is that GMM's soft assignments collapsed to hard 0/1 probabilities, providing no uncertainty quantification. With only 20 observations and 15 features, the GMM covariance matrices are severely under-constrained, causing posterior responsibilities to become near-deterministic. This is a known consequence of high feature-to-observation ratios [3].

5.5 Visualisation

Fig. 3 presents 3D scatter plots of possession (`Poss`), shots (`Sh`), and tackles won (`TklW`) for both methods. These three features capture the key style dimensions: attacking intent (shots), territorial dominance (possession), and defensive work rate (tackles). The elite cluster lies at the top-right (high possession, high shots), while the struggling cluster sits at the bottom-left. The mid-table group occupies the centre but with notably higher tackle rates, reflecting a reactive defensive style.

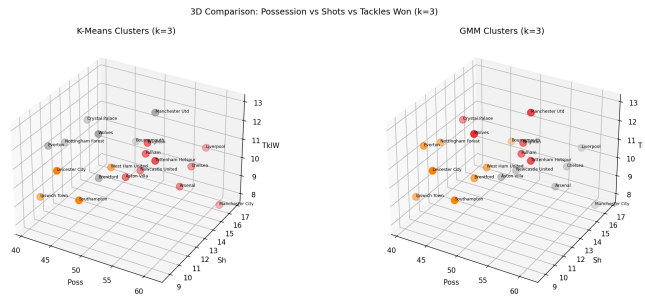

Figure 3: 3D scatter: Possession vs Shots vs Tackles Won for K-Means (left) and GMM (right), both with $k = 3$.

6 Concluding Remarks

6.1 Achievements

This analysis demonstrates that meaningful, interpretable playing style clusters exist within EPL squad statistics. Both K-Means and GMM independently identify a core attacking group (Arsenal, Aston Villa, Chelsea, Liverpool, Manchester City, Newcastle United) and distinguish it from the remaining sides. The consistency of this grouping across two methodologically distinct algorithms strengthens confidence in the finding. Cluster profiles align with tactical intuition: attacking-style teams dominate possession and generate more shots; defensive-style teams win the most tackles; balanced teams occupy the middle ground on most metrics.

6.2 Limitations

- **Small sample size:** With only 20 observations, the dataset is at the lower boundary for meaningful clustering. GMM's soft-assignment advantage was entirely lost: the 15:20 feature-to-observation ratio caused all posterior probabilities to collapse to 0 or 1.

- **Single season:** Playing styles evolve with managerial changes, injuries, and squad invest-

ment. A single season may not capture stable tactical identities, particularly for clubs undergoing transitions.

- **Squad-level aggregation:** Averaging across a full season discards match-to-match and player-level variation. Teams with high game-to-game tactical flexibility are misrepresented by their season means.
- **No opponent context:** Defensive metrics (`Int`, `TklW`) depend partly on the opponent's attacking intent. A team facing many weak opponents may record fewer interceptions independently of their defensive organisation.

6.3 Future Directions

- **Multi-season data:** Extending to 3–5 seasons increases the effective sample size and enables temporal analysis of style evolution.
- **Player-level clustering:** Clustering individual players unlocks far larger datasets (500+ players per season), removing the sample size constraint on GMM's probabilistic advantages.
- **Opponent-adjusted statistics:** Using opponent-adjusted defensive metrics would isolate team-level tendencies from opponent quality effects.
- **Wider leagues:** Including multiple European leagues would increase the sample to 80–100 teams, making GMM's full probabilistic framework realisable.

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